ModelIt!

Student Activity Pack

The Invisible Force That Charges Your Phone

How Electricity and Magnetism Shape Our World

Grade 7 | NGSS MS-PS2-3, MS-PS2-5 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is a magnet and how does it work?

2 Can you make a magnet stronger or weaker? How?

3 What do you think connects electricity and magnetism?

4 Draw what you think a magnetic field looks like around a magnet.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Electric Current | Number of Coil Wraps | Magnetic Field Strength | Lifting Capacity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Low Power

Low current, few coils

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Maximum Coils

Many coils, moderate current

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Full Power

Maximum current AND maximum coils

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Core Material — The type of metal inside the coil — iron cores concentrate magnetic fields better than air

Distance from Object — Magnetic force decreases rapidly with distance — the closer the object, the stronger the pull

Temperature — Heating a magnet weakens it because thermal energy disrupts the alignment of magnetic domains

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a force you can't see, smell, or touch move objects and transfer energy? Use evidence from your simulation to support your answers.

1. **Why are electromagnets more useful than permanent magnets in many applications?**

2. **How does your phone use electromagnetic forces every day?**

3. **What would happen to our technology if we couldn't create magnetic fields with electricity?**

4. **Why does the magnetic force get weaker when you move farther away?**

5. **How does your model prove that invisible forces are real even though we can't see them?**

8 STEM Challenge: Design the Ultimate Electromagnet

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YOUR ENGINEERING MISSION

Engineer the strongest possible electromagnet using limited materials, then test it by measuring how much weight it can lift.

THE SCENARIO

A recycling facility needs a new electromagnet crane to sort metal from waste. Your engineering team must design and test a prototype that lifts the maximum weight using the materials provided.

Engineering Design Thinking

1. HOW MANY COIL WRAPS WILL YOU USE AND WHY?
2. WHAT POWER LEVEL WILL YOU SET YOUR CURRENT TO?
3. WHAT CORE MATERIAL WILL GIVE THE BEST RESULTS?
4. HOW WILL YOU MEASURE AND RECORD YOUR ELECTROMAGNET'S LIFTING CAPACITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Electrical Engineers design circuits, motors, generators, and electromagnetic devices used in everything from phones to MRI machines. They earn $75,000–$130,000/year.